

Electric field web simulation

<https://mariododo.github.io/electric-field-simulator/>

Střední škola Teleinformatiky, Ostrava, p.o.

1st Mário Daniel Doskočil
Karviná, Czech Republic
mddoskocil@pm.me

2nd Monika Pustková
Opava, Czech Republic
monika@pustkova.cz

3rd Martin Holánik
Ostrava, Czech Republic
martin.holanik06@gmail.com

Abstract—This paper describes a web-based simulation of an electric field primarily designed for interactive lessons in lower education.

Index Terms—Simulation, HTML5, Web, Svg, Javascript, Coulomb's law

I. INTRODUCTION

The goal of this project is to create a teaching tool on the theme of "Electric Field," primarily for physics classes. The tool displays electric charges and their corresponding interactions using field lines, compass needles, or animations, enabling students to visualize and understand the concepts easily. User-selectable parameters, such as the intensity, length, and quality of field lines, or the colors of all rendered elements, allow for customization of the simulation. The simulation result can be exported to an SVG file for further use. The entire simulation is developed using JavaScript, without any external libraries. Moreover, the calculations used for the simulation can also be applied in other related applications.

II. DESCRIPTION

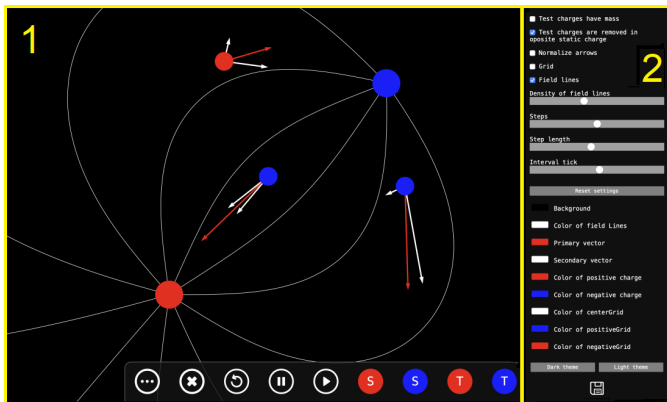


Fig. 1. View of the simulation

A. Basic view

In the basic view, you can see a black background with a bar at the bottom of the screen. On the bar, there are buttons (in order from left): advanced menu, remove all, restart, stop, and

play. Additionally, static and test charges of opposite charge are in different colors.

B. Work with website:

1) *Addition of charges:* To add a charge, drag it from the bar to the black area. Field lines will appear around positive charges, while force vectors will be displayed around test charges in the direction of static charges of opposite charge. The resultant force is displayed as a red vector.

2) *Removal of charges:* You can remove the charges by dragging them back to the bar. Additionally, charges will be automatically removed once they reach 200 pixels behind the black area in every direction.

3) Action buttons:

- Play - Starts the animation
- Pause - Stops the animation
- Reset - Returns the charges to their previous position
- Remove all - Removes all elements from the black area
- Advance settings - Opens advance settings menu

C. Advanced settings

- Test charges have mass - Enables test charges to act on the field
- Test charges are removed in opposite static charge
- Normalize arrows - Gives the same length to all vectors
- Grid - Toggle the compass needle grid
- Field Lines - Toggle field lines
- Density of fields lines - Amount of the initial field lines points
- Step length - Length of one field line rendering step
- Interval tick - Sets delay for the interval that drives animation
- Reset settings - Resets all settings to default values
- Change the color of background, field lines, vectors, charges and needle grid
- Dark/Light - Sets dark or light theme (for example for projection on a board)
- Save as SVG - Saves simulation to the .svg format

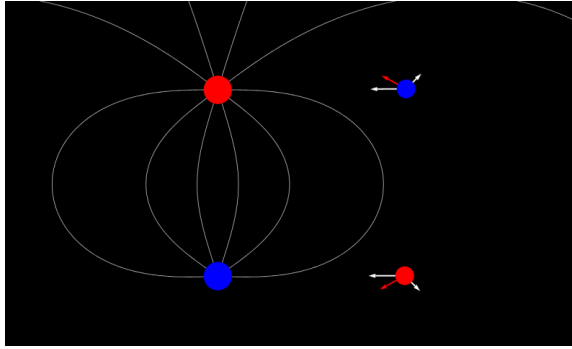


Fig. 2. Default settings view

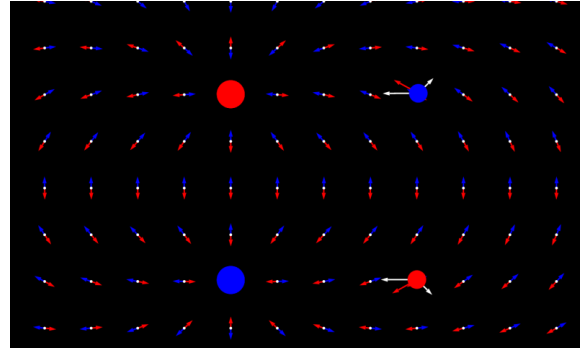


Fig. 6. Compass needle grid without the field lines

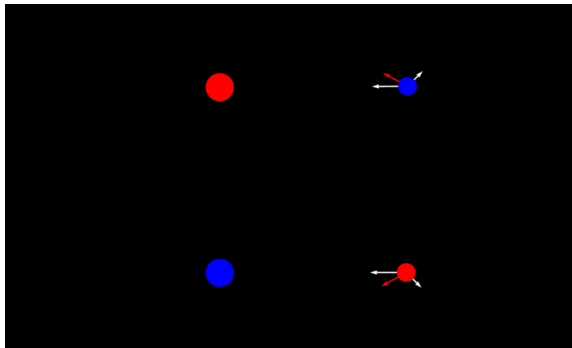


Fig. 3. Field lines and the needle grid disabled

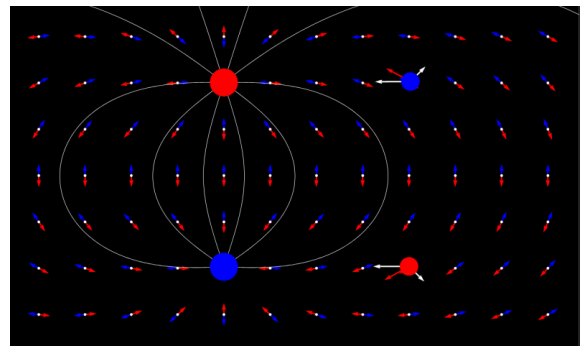


Fig. 7. Compass needle grid with the field lines

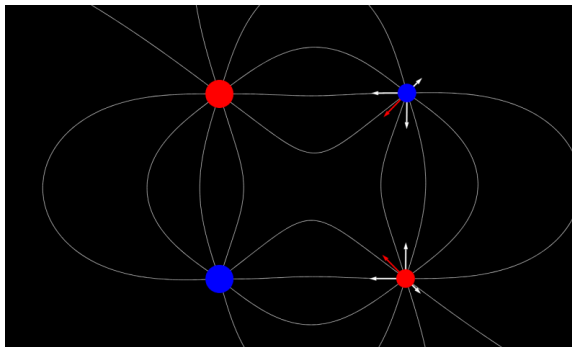


Fig. 4. Test charges have mass

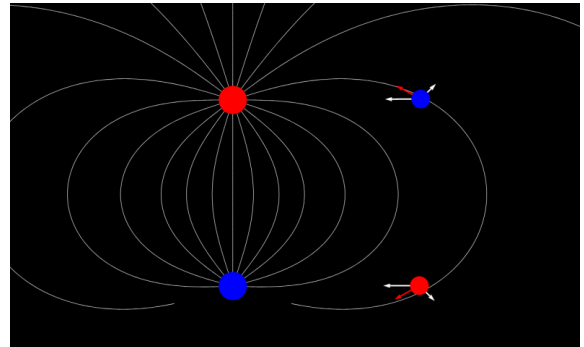


Fig. 8. Step length set to the maximum value

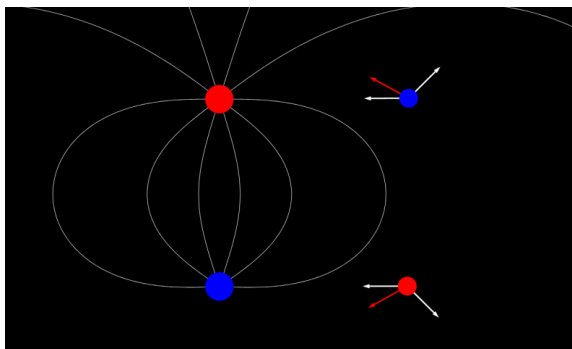


Fig. 5. Normalized vectors

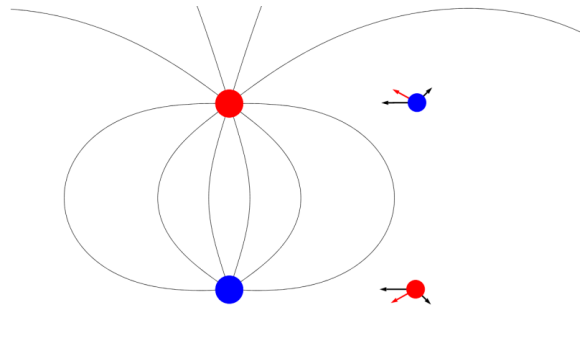


Fig. 9. Light mode

III. ALGORITHMS

Charges Q are objects with the following attributes:

- x - The x coordinate of the charge on the screen.
- y - The y -coordinate of the charge on the screen.
- N - A boolean value that determines the sign of the charge.

Calculations involving angles are performed using radians because the standard JavaScript Math library utilizes them.

A. Calculating force

To calculate the force between two charges, Coulomb's law is used.

$$F = k_e \cdot \frac{|Q_1| \cdot |Q_2|}{r^2}$$

The Coulomb's law is used in a modified form where $k_e = 1$ and $Q_1, Q_2 = 1000C$ which results in:

$$F = \frac{10^6}{r^2}$$

B. Calculating the distance r

To calculate the distance between two charges, we can use the Pythagorean theorem. To calculate the overhangs, we have to take the absolute value of the difference between their respective coordinates.

$$x = |Q_{1x} - Q_{2x}|$$

$$y = |Q_{1y} - Q_{2y}|$$

$$r = \sqrt{x^2 + y^2}$$

C. Resultant force on a charge

The end coordinates of the resultant force U_r can be calculated by summing up all the individual end coordinates U_p of the forces acting on a charge.

$$U_r = \begin{bmatrix} U_{rx} \\ U_{ry} \end{bmatrix} = \begin{bmatrix} \sum_{i=0}^{f-1} U_{pix} \\ \sum_{i=0}^{f-1} U_{piy} \end{bmatrix}$$

Where f is the number of forces acting on a specific charge.

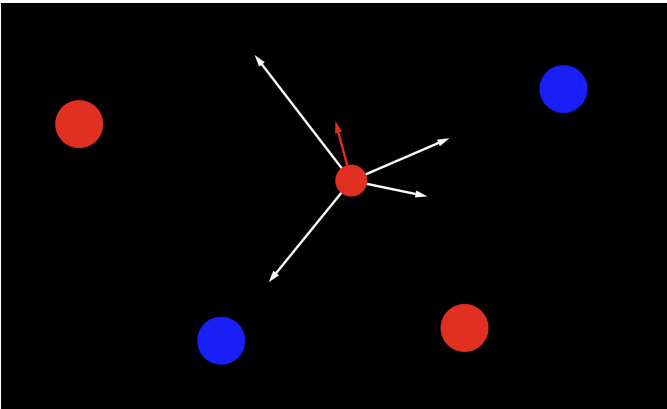


Fig. 10. The white arrows represent the individual forces acting on a charge, while the red arrow represents the resultant force.

D. Generating field lines

The number of field lines n is determined by the user. To start drawing the field lines around a positive charge, we need to distribute the initial points I of the field lines around the charge.

$$\alpha_i = 2\pi - i \cdot \frac{2\pi}{n}$$

$$I_i = \begin{bmatrix} I_{ix} \\ I_{iy} \end{bmatrix} = \begin{bmatrix} Q_{ix} + r \cdot \cos(\alpha_i) \\ Q_{iy} + r \cdot \sin(\alpha_i) \end{bmatrix}$$

Where $i \in \mathbb{N}_0 \wedge 0 \leq i < n$ and r is the radius of the charge. Then we can calculate the force at each initial point and normalize the vector representing the force, so that we obtain the direction of the next step for the field line.

$$u_n = \frac{u}{|u|}$$

But since we are interested in specific coordinates, we can rewrite the formula as follows, where λ is the length of one field line rendering step:

$$L_i = \begin{bmatrix} L_{ix} \\ L_{iy} \end{bmatrix} = \lambda \begin{bmatrix} \frac{I_{ix}}{\sqrt{I_{ix}^2 + I_{iy}^2}} \\ \frac{I_{iy}}{\sqrt{I_{ix}^2 + I_{iy}^2}} \end{bmatrix}$$

We will get a new point L_i , which is the end coordinate of one field line step.

Then, we set I_i to be equal to L_i and repeat the whole process for the user-defined number of steps.

E. Needle grid

For this type of visualization, we have to create a 2D matrix of points.

$$G = \begin{bmatrix} G_{00} & \cdots & G_{0j} \\ \vdots & \ddots & \vdots \\ G_{i0} & \cdots & G_{ij} \end{bmatrix}$$

$$i, j \in \mathbb{N}_0 \wedge 0 \leq i < \frac{\text{screen-height}}{100} \wedge 0 \leq j < \frac{\text{screen-width}}{100}$$

Each point G_{ij} has a default offset of 20 pixels from the screen border. The gaps between the needles are 100 pixels.

$$G_{ij} = \begin{bmatrix} 20 + j \cdot 100 \\ 20 + i \cdot 100 \end{bmatrix}$$

For each point G_{ij} , we have to calculate the force acting on it, normalize that force to get the direction of the needle, and calculate the end coordinate of the needle.

To create the second needle in the opposite direction, we take each end coordinate of the first needle and replace it with its opposite value.

IV. FEEDBACK AND USE IN PRACTICE

During the realization of the project, the teachers gave us feedback. Originally, the application was only in dark colors. One of the problems was that it was poorly visible during projection. That's why we added an option to customize the colors of all objects. For quick changes, we also added a light and dark mode button.

We have made the final application available to teachers of electrical engineering and physics, who were happy that we were interested in the topic within their specialization. The final version of the application has already been used in electrical engineering classes.

According to teachers, the application is interesting for students as they do not only see the subject from the point of view of drawn lines on the blackboard, but also observe how the charges dynamically interact with each other.

For future updates we would like to add a calculation of the voltage between charges. The higher the voltage, the brighter the color of the background will be. Additionally, we plan to include the dependence of the voltage on the environment, such as in a vacuum, air, water, etc.

V. CONCLUSION

We have developed an educational tool that has proven to be practical and effective in enhancing the learning experience of students. This tool is freely accessible to the public, and its source code is also available, making it easily modifiable. Our project has already been utilized in real-world settings and has been evaluated as successful in achieving its objectives.

VI. ACKNOWLEDGMENT

Thanks to our supervisor, Ing. František Knápek, for his patience and time he dedicated to us. Special thanks also go to our electrical engineering teacher, Ing. Václav Jedlička, for introducing us to this topic and explaining it. We are glad that he is using our tool in his classes and we believe that it helps students understand the electric field lessons better.